Corn leaf diseases: Gray leaf spot vs. Northern Leaf Blight

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Question: is it plausible that the corn leaves our model confuses carry both diseases at once? **Answer: for some of them, not all.** 4 of the 6 wrong test images (#5, #7, #11, #12) show a separate NLB-like and GLS-like area, and the test images are more mixed than validation leaves ($p = 0.02$). Two errors (#6, #8) look like GLS only, which fits a mislabel better.

Results

A 21-class EfficientNetB0 trained on the PlantVillage color images reaches 97.8% validation accuracy and 97.8% on the team's 10,709-image test split, but gets only 8 of the 14 held-out test images right. All six errors are swaps between Northern Leaf Blight (NLB, *Exserohilum turcicum*) and Gray leaf spot (GLS, *Cercospora zeae-maydis*): four NLB leaves called GLS and two GLS leaves called NLB.

- **Data:** 54,305 images, labelled by disease only (21 classes). The 14 test images from `Guess.txt` and the 10,709 images of the team's test split (`test.txt`) were excluded from training; 10% of the remaining 43,582 was held out for validation (4,359 images).
- **Among the weakest classes:** GLS (F1 0.91) is the second-lowest class after Target Spot (0.85); NLB scores 0.95. On the test split GLS reaches 91.6% and NLB 94.5%, and 20 of their 21 errors are the other corn disease.
- **Not a data-volume problem:** GLS has 399 training and validation images and NLB 758, more than Cedar apple rust (223), which scores 100% on the test split.
- **The model sees a two-way choice:** on 13 of the 14 test images the top two guesses are GLS and NLB, together 97.9% or more. The exception is #14, a confident GLS (94.3%) with Apple scab second (3.1%).
- **Small test set:** with 14 images, 8/14 fits a true accuracy anywhere from about 33% to 79% (95% interval).

Predictions on the 14 test images

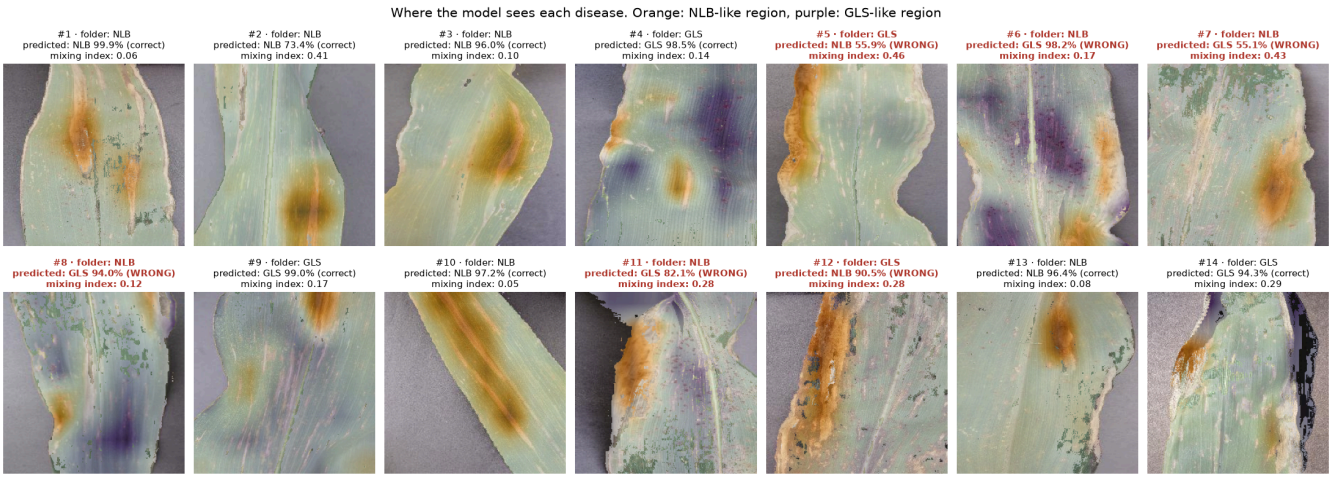
#	Folder label	1st guess	2nd guess	3rd guess	Correct?
1	NLB	NLB 99.9%	GLS 0.1%	Late blight 0.0%	yes
2	NLB	NLB 73.4%	GLS 26.6%	Common rust 0.0%	yes
3	NLB	NLB 96.0%	GLS 4.0%	Common rust 0.0%	yes
4	GLS	GLS 98.5%	NLB 1.5%	Bacterial spot 0.0%	yes
5	GLS	NLB 55.9%	GLS 42.0%	Bacterial spot 2.0%	**no**
6	NLB	GLS 98.2%	NLB 1.8%	Common rust 0.0%	**no**
7	NLB	GLS 55.1%	NLB 44.9%	Common rust 0.0%	**no**
8	NLB	GLS 94.0%	NLB 5.9%	Powdery mildew 0.0%	**no**
9	GLS	GLS 99.0%	NLB 1.0%	Common rust 0.0%	yes
10	NLB	NLB 97.2%	GLS 2.7%	Late blight 0.0%	yes
11	NLB	GLS 82.1%	NLB 16.5%	Common rust 1.3%	**no**
12	GLS	NLB 90.5%	GLS 9.1%	Leaf scorch 0.2%	**no**
13	NLB	NLB 96.4%	GLS 3.5%	Common rust 0.0%	yes
14	GLS	GLS 94.3%	Apple scab 3.1%	Common rust 1.6%	yes

Confidences are averaged over the original image and three flipped copies. The folder label is the dataset's label and was used only for scoring, never shown to the model.

Why the model confuses them

Most misclassified leaves show two separate regions, one NLB-like and one GLS-like. That fits co-infection, which single-label PlantVillage cannot represent.

The model's last step averages an 8×8 feature grid and then scores it. That lets us split each disease score exactly by leaf region (a class activation map). From the per-region difference between the NLB and GLS scores we compute a **mixing index**: 0 when one disease dominates the whole leaf, 0.5 when both are equally present in different places.



Above each leaf: its folder label, the model's top guess with confidence (red = wrong), and its mixing index.

The table below uses the **validation set**, not the 14 test images. It holds 116 corn images (40 GLS + 76 NLB), grouped two ways: by folder label, and by whether the model classified them correctly. The model got 109 right and 7 wrong.

Validation corn images (116 in total)	Images	Median mixing index
By folder: GLS	40	0.17
By folder: NLB	76	0.08
By prediction: correct	109	0.09
By prediction: misclassified	7	**0.18**

- **One-sided mixing:** 45% of GLS-labelled validation leaves have an NLB-like region (index > 0.2), against 18% of NLB leaves.
- **Errors happen on more mixed leaves:** the 7 misclassified validation images have twice the median mixing index of the 109 correct ones. With only 7 errors this is a weak signal.
- **The test images are more mixed than validation:** median 0.17 vs 0.10 ($p = 0.02$), and the 6 wrong ones more so ($p = 0.004$).
- **The 6 wrong test images are not all alike:**
  - #5 (0.46) and #7 (0.43) show a clear NLB-like lesion plus a separate GLS-like area, close to the most mixed of all 130 corn images (0.48): plausibly both diseases.
  - #11 (0.28) and #12 (0.28) have both regions, one clearly larger: possible.

    - #6 (0.17) and #8 (0.12) are GLS-like on about 80% of the leaf despite the NLB label: more likely mislabels or GLS-only leaves.
- **Some correctly classified test leaves look mixed too:** #2 (0.41) and #14 (0.29).

Conclusion and limitations

Co-infection is a plausible explanation for some GLS/NLB confusions (#5, #7), and the test images are measurably more mixed than validation leaves. It doesn't explain every error: #6 and #8 look like label noise. This is supporting evidence, not proof.

- **"GLS-like" only means similar to the GLS training images.** Other spots, such as rust, could contribute to it.
- **Small sample:** the test images differ from validation at p = 0.02, but that rests on 14 images and only 7 validation errors.
- **Coarse maps:** 8×8 cells of 32×32 pixels, so small spots aren't resolved individually.
- **Model-dependent:** the index shows what this model learned to associate with each disease, not a direct measurement of the leaf.
- **Confirmation needs an expert:** a plant pathologist would have to examine the leaves, or the data would need multi-label annotation.

Suggested wording for the report:

> Class activation maps show that most misclassified corn leaves contain spatially separate regions of Northern Leaf Blight and Gray leaf spot evidence (median mixing index 0.18 vs 0.09 for correctly classified validation leaves), and the 14 test images are more mixed than the validation set (p = 0.02). This is consistent with co-infection, which PlantVillage's single-label annotation cannot represent. Some errors, such as test images #6 and #8, show uniform GLS evidence instead, consistent with label noise.

How this was made

- **AI assistant:** analysis, code and write-up by Claude Opus 5.5 (model ID `claude-opus-5-5`, Anthropic), running in Claude Code in VS Code.
- **Training:** TensorFlow 2.15.1 on an NVIDIA RTX 3060 Laptop GPU (6 GB); EfficientNetB0 pretrained on ImageNet, trained for 4 head epochs, then fine-tuned. The loss diverged in fine-tuning epoch 4, so early stopping kept the epoch-3 weights.

- **Scripts:** `train_no_test.py` (21-class model and test predictions), `eval_test.py` (results on `test.txt`), `coinfection_notest.py` (activation maps and mixing index).